

Comprehensive Markdown Example

Table of Contents

- Introduction
- Typography
- Lists
- Links and Images
- Blockquotes
- Code and Syntax Highlighting
- Tables
- Task Lists
- Footnotes
- Diagrams (Mermaid)
- HTML
- Mathematics (KaTeX)
- Conclusion

Introduction

This document demonstrates **comprehensive Markdown syntax** using GitHub Flavored Markdown (GFM) extensions, plus common extensions like footnotes, task lists, tables, task lists, diagrams via Mermaid, and mathematics via KaTeX/MathJax.

Note: This document showcases features available in GitHub Flavored Markdown and common Markdown extensions.

Typography

- *Italic*, **bold**, `code`, ~~strikethrough~~, **==highlight==**, ^{superscript}, _{subscript}
- ***Bold and italic***, **`**bold code**`**
- Links with title attribute
- Reference-style link
- Reference-style link

Lists

Unordered

- Item 1
- Item 2
 - Nested item A
 - Nested item B
- Item 3

Alt text

Figure 1: Alt text

Alternate text without title

Figure 2: Alternate text without title

Ordered

1. First item
2. Second item
 1. Nested ordered
 2. Nested ordered
3. Third item

Mixed

- Task lists (see below)
- Definition lists (via HTML or pandoc extensions)

Definition Lists (via HTML or pandoc-style)

Term 1

Definition 1

Term 2

Definition 2

Or using pandoc-style:

Term 1 Definition 1

Term 2 Definition 2

Links and Images

Alt text

Blockquotes

This is a blockquote.

This is a nested blockquote.

Alt text

Figure 3: Alt text

Back to level 1.

- A list inside a blockquote
- Another item

Code and Syntax Highlighting

Inline code has backticks around it.

```
var s = "JavaScript syntax highlighting";
alert(s);

def fibonacci(n):
    """Return the nth Fibonacci number."""
    a, b = 0, 1
    for _ in range(n):
        yield a
        a, b = b, a + b

<!DOCTYPE html>
<html>
<head>
    <title>Example</title>
</head>
<body>
    <h1>Hello, world!</h1>
</body>
</html>

SELECT *
FROM users
WHERE age > 18
ORDER BY name;
```

Tables

Left-aligned	Center-aligned	Right-aligned
Cell 1	Cell 2	Cell 3
Cell 4	Cell 5	Cell 6
Very long text that wraps	More text	Even more text

Header 1	Header 2	Header 3
Row 1 Col 1	Row 1 Col 2	Row 1 Col 3
Row 2 Col 1	Row 2 Col 2	Row 2 Col 3

Task Lists

- ☒ Write the press release
- ☐ Update the website
- ☐ Contact the press
- ☐ **Done!**
- ☒ **Completed task**
- ☐ Incomplete task
 - ☐ Nested sub-task
 - ☒ Completed sub-task

Footnotes

Here's a sentence with a footnote. ¹

Here's another footnote reference.²

Diagrams (Mermaid)

```
graph TD;
  A[Start] --> B{Is it working?};
  B -->|Yes| C[Done];
  B -->|No| D[Try Again];
  D --> B;

sequenceDiagram
  Alice->>Bob: Hello Bob, how are you?
  alt is sick
    Bob->>Alice: Not so good :(
  else is well
    Bob->>Alice: Feeling good like always :)
  end
  Alice->>Bob: Glad to hear it.

pie
  title Pets adopted by volunteers
  "Dogs" : 386
  "Cats" : 85
  "Rats" : 15
```

¹This is the first footnote.

²Here's one with multiple blocks.

Subsequent paragraphs are indented to show that they belong to the previous footnote.

```
{ some.code }
```

The whole paragraph can be indented, but the first line must be spaced correctly.

```
{ more.code }
```

HTML

You can also use raw HTML in Markdown (depending on the renderer).

Definition list

Is something people use sometimes.

Markdown in HTML

Does *not* work **very** well. Use HTML tags.

Name

Age

Alice

25

Bob

30

Mathematics (KaTeX/MathJax)

Inline math: $E = mc^2$.

Display math:

$$\frac{d}{dx} \left(\int_0^x f(u) du \right) = f(x).$$

$$\nabla \times \vec{\mathbf{B}} - \frac{1}{c} \frac{\partial \vec{\mathbf{E}}}{\partial t} = \frac{4\pi}{c} \vec{\mathbf{j}}$$

$$\nabla \cdot \vec{\mathbf{E}} = 4\pi\rho$$

$$\nabla \cdot \vec{\mathbf{B}} = 0$$

$$\nabla \times \vec{\mathbf{E}} + \frac{1}{c} \frac{\partial \vec{\mathbf{B}}}{\partial t} = 0$$

Conclusion

Markdown is a versatile markup language that, with extensions, can handle most documentation needs.